

GIZMO Operations Guide

Never lift the safety ground. Follow SOP004 and the current work plan.

Installed system

GIZMO is already deployed. SOP004 authorizes its installation and use. The initial grounding setup will use one saturable inductor.

Item	Recorded setting
Excitation	1436 Hz
Alarm threshold	100 Ω
Runtime / OPC UA model	0.5.1 / 1.4.0
Monitoring server	dune-fd-test01
History available	Since July 27, 2026

These are stored settings, not live confirmation. As of September 30, 2026, the instrument is unreachable from the monitoring server. The latest retained sample is September 13, 2026, at 07:11:34 UTC. The server dashboard and historian are running.

Connections after installation or maintenance

With the rack de-energized, check each cable against its installed label and the approved drawing:

- Pearson CT output to GIZMO CT input; the CT observes the intended safety-ground path.
- Stimulus center to the designated Detector Ground point; return to the designated rack/infrastructure reference.
- Beacon and audible output to their labeled connectors.
- Network to the assigned controls port; approved DC supply to the chassis.
- One saturable inductor in the initial setup, with its protection and safety-ground connections verified by the electrical lead.

Changes to grounding, CT routing, stimulus, or calibration require the applicable work controls. Do not improvise a bypass or move connections to clear an alarm.

Contractor shift checks

Contractor personnel work with the cryostat and beam structure connected to Building Ground. At shift start and handover, acknowledge the grounding status:

1. Check the Building Ground bond with a multimeter.
2. Check the knife-switch position and tagout status.
3. Clear tools and electrical cords so the FNAL I&I/GIZMO group can test before, after, or between shifts.

GIZMO readings, dashboard checks, and isolation tests are the responsibility of the FNAL I&I/GIZMO group.

FNAL I&I/GIZMO isolation tests

Use the planned knife-switch/tagout arrangement for periodic tests of shorts, capacitance, and coupling between Detector Ground and Building Ground.

1. Coordinate the test window with the contractor and confirm the area is clear of tools and electrical cords.
2. Establish and record the test grounding state using the authorized switching/tagout procedure.
3. Check fresh data, measurement quality, alarm, frequency, and threshold. Record isolation, valid capacitance, and coupling diagnostics; compare with the permitted limits.
4. Restore and verify the required Building Ground connection, complete tagout closeout, and hand the structure back before contractor work resumes.

Plan tests before, after, or between shifts, often enough to avoid disruptive late discovery of unwanted connections. Test cadence, detailed shift routine, settling time, permitted isolation limits, and failed/missed-test response: TBD. The recorded 100 Ω alarm threshold is not an approved construction-isolation acceptance limit.

Record test time, grounding state, findings, and handback in Procore Daily Log > GIZMO. HIGH Z is a range

indication, not a resistance value. The last stored sample had Uncertain quality, an uncertain clear alarm, and a latched alarm; it does not establish the present ground condition.

FNAL: alarm response

1. Check the recorded grounding state and any intentional construction or welding bond. An isolation result is not meaningful while the required Building Ground connection is in place.
2. If the alarm is unexpected for the recorded grounding state, stop new ground-affecting work. Record the time and alarm reason; notify the activity coordinator and grounding/GIZMO contact.
3. Check what changed: welding returns, temporary leads, cable shields, piping, scaffolding, or tools. Investigate under the work plan.
4. Keep the data and record the disposition. Do not change the threshold to make the alarm disappear.

FNAL: lost data

Treat stale or non-Good data as unavailable monitoring. A running dashboard or an old clear alarm is not proof that GIZMO is working.

1. Record the last sample time and notify the GIZMO and SC/DPS contacts.
2. During an isolation test, do not accept a result without valid data or an approved alternative. During contractor work, verify the required Building Ground connection; continuation and rescheduling follow the work plan.
3. Check the instrument power and network through the responsible on-site contact. The server reports loss of reachability; the cause has not been established.
4. After connection returns, verify advancing timestamps, usable measurement/alarm quality, and resumed history before relying on GIZMO again.

Welding

Use the approved welding-return arrangement. The FNAL I&I/GIZMO group manages alarm inhibition and retains alarms and history. Restore the contractor grounding state after work; the FNAL group checks isolation at the scheduled test.

FNAL: server check

On dune-fd-test01, run:

```
/storage/gizmo-monitor/bin/status
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It reports the historian, dashboard, upstream connection, stored samples, and disk space. Check upstream connectivity and the newest GIZMO sample, not just whether the processes are running.

For a local dashboard connection:

```
ssh -N -L 127.0.0.1:18081:127.0.0.1:18081 dune-fd-test01
```

Then open <http://127.0.0.1:18081/> in the browser. Use the configured SSH access for the server.

FNAL: shutdown

Coordinate any loss of monitoring before shutdown. Use SOP004 and the deployed operating procedure to disable excitation and power down. Leave the permanent safety-ground path intact. Record the time and reason.